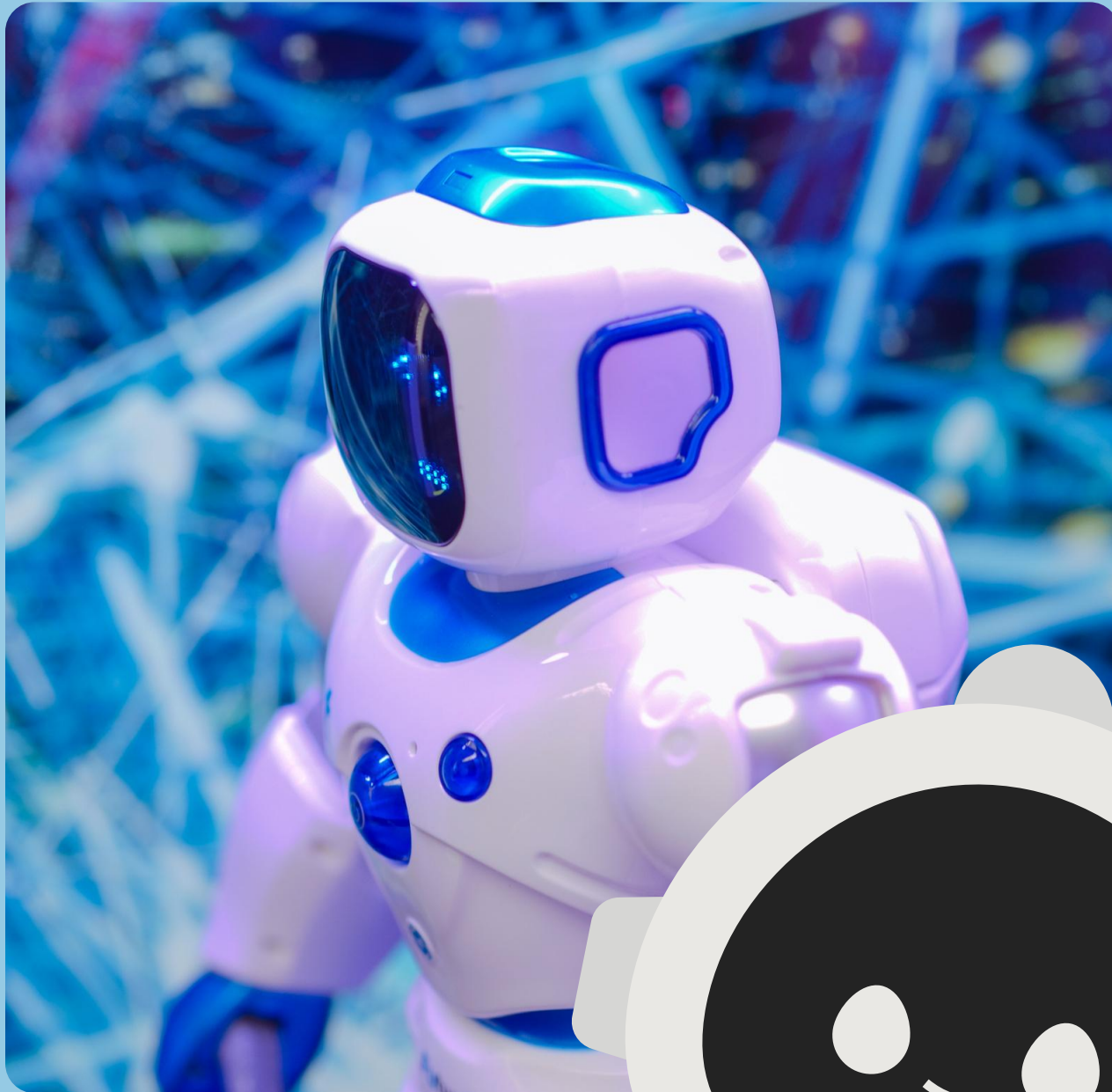




InvestorGPT

AIMP: Advanced LLM for the
Manufacturing Industry @ PSDC

Presented by
Group 3 Capstone Project Team

Boon Ching

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What is InvestorGPT ?

Investment research is fragmented across long reports, live prices, technical indicators and multiple websites.

CORE VALUE: one interface for documents, analytics, forecasts and evidence



Challenges

- Annual reports are lengthy
- Numeric and narrative data are separated
- Manual comparison takes time
- Unsupported AI answers can hallucinate

Objective

- Ask natural-language questions
- Retrieve evidence from source documents
- Compute stock metrics directly from CSV
- Compare Intel and Micron visually

Success Criteria

- Grounded, citation-backed answers
- Accurate financial calculations
- Multi-company comparison
- Working end-to-end prototype

What the Prototype Delivers?

2028

Static HTML

Static + AI Chat

Streamlit Live

01

AI chat with citations

Natural-language Q&A grounded in retrieved financial evidence

02

Multi-source RAG

SEC filings and financial web content indexed in FAISS

03

Structured analytics

Return, CAGR, volatility, drawdown and moving averages from CSV

04

Investment dashboard

Price, signal, Buffett score, fundamentals, forecast and comparison

ACTION
Decision Matrix
Quality × Timing

BUY

DCA

HOLD

REDUCE

SELL

05

Intelligent routing

Question is directed to CSV, FAISS, mixed or conversational path

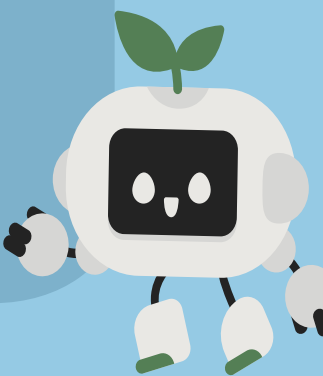
06

Session memory

Conversation history is retained within browser sessions



InvestorGPT



INVESTORGPT ARCHITECTURE

Retrieval-Augmented Generation Financial Research System

RAG + Intelligent Routing Citations & Sources Memory & Context Analytics & Insights

INTELLIGENT QUERY ROUTING

- csv_price Stock price lookup
- csv_analytics Financial analytics
- faiss Document retrieval
- mixed CSV + Documents
- conversational Memory only

ENTITY EXTRACTION

- Tickers INTC, MU
- Years 2019-2026
- Intent What, Why, How

STRUCTURED MARKET DATA PATH

STOCK DATA

OHLCV CSV
INTC, MU

FINANCIAL ANALYTICS

- Total Return
- CAGR
- Volatility
- Drawdown
- Moving Averages
- & More

DOCUMENT RETRIEVAL PATH

EMBED QUERY

text-embedding-3-large (3072-dim)

RETRIEVE TOP-K

Top 8 chunks
Metadata filters

RANK & FILTER

Confidence scoring
Low-score filter

LLM SYNTHESIS

- GPT-4o (Chat Completion)
- Context + History + Instructions
- Grounded Answers with Citations

[1] Intel 10-K FY2024 (p. 42)
[2] Micron IR - Press Release
[3] Macrotrends - Revenue Data

VECTOR DATABASE

FAISS IndexFlatIP
754 vectors
Cosine Similarity
Metadata Rich

DATA & STORAGE LAYER

STOCK HISTORY CSV

INTC: 11,691 rows (1980-2026)
MU: 10,626 rows (1984-2026)

CRAWLED MARKDOWN

21+ Sources
Markdown + Metadata JSON

10-K HTML FILINGS

6 Filings
Intel & Micron
FY2022-FY2025

FAISS VECTOR INDEX

754 Chunks
3072-dim
Cosine Similarity

CACHE & METADATA

.cache/
Refresh Metadata
Timestamps

SESSIONS & MEMORY

In-Memory
1-week TTL
100 msgs/session

EXTERNAL DATA SOURCES



SEC EDGAR API
10-K Filings
Intel & Micron



FIRECRAWL SDK
Web Scraping
21+ Sources



YAHOO FINANCE API
Stock Prices
OHLCV Data



MACROTRENDS
Financial Metrics
10Y History



STOCKANALYSIS
Annual & Quarterly
Financials



COMPANIES MARKETCAP
Market Cap
History



INVESTOR RELATIONS
Press Releases
Presentations

SYSTEM SERVICES



FLASK API SERVER
Port 8503



SESSION MEMORY
Context & History



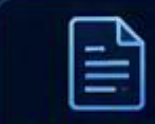
HEALTH MONITOR
/health Check



QUERY ROUTER
Intent Detection



ANALYTICS ENGINE
CSV Metrics



EXPORT ENGINE
Reports & Data

USER INTERFACE



- Ask questions
- Analyze companies
- Compare metrics
- Get insights

INTERFACES

STREAMLIT DASHBOARD



Port 8502

STATIC HTML DASHBOARD



AI Chat Tab

OUTPUT & INSIGHTS

- AI Answers with Citations
- Financial Analytics
- Comparisons & Benchmarks
- Charts & Visualizations
- Decision Matrix & Signals
- Export & Reports



→ User Flow - - - Data Ingestion → Structured Data Flow → Retrieval Flow → Synthesis Flow 🗉 Citation Output

Architecture

How RPA Works?

2028

754

FAISS vectors

3,072

Embedding dimensions

Chunking strategy

400–800 tokens

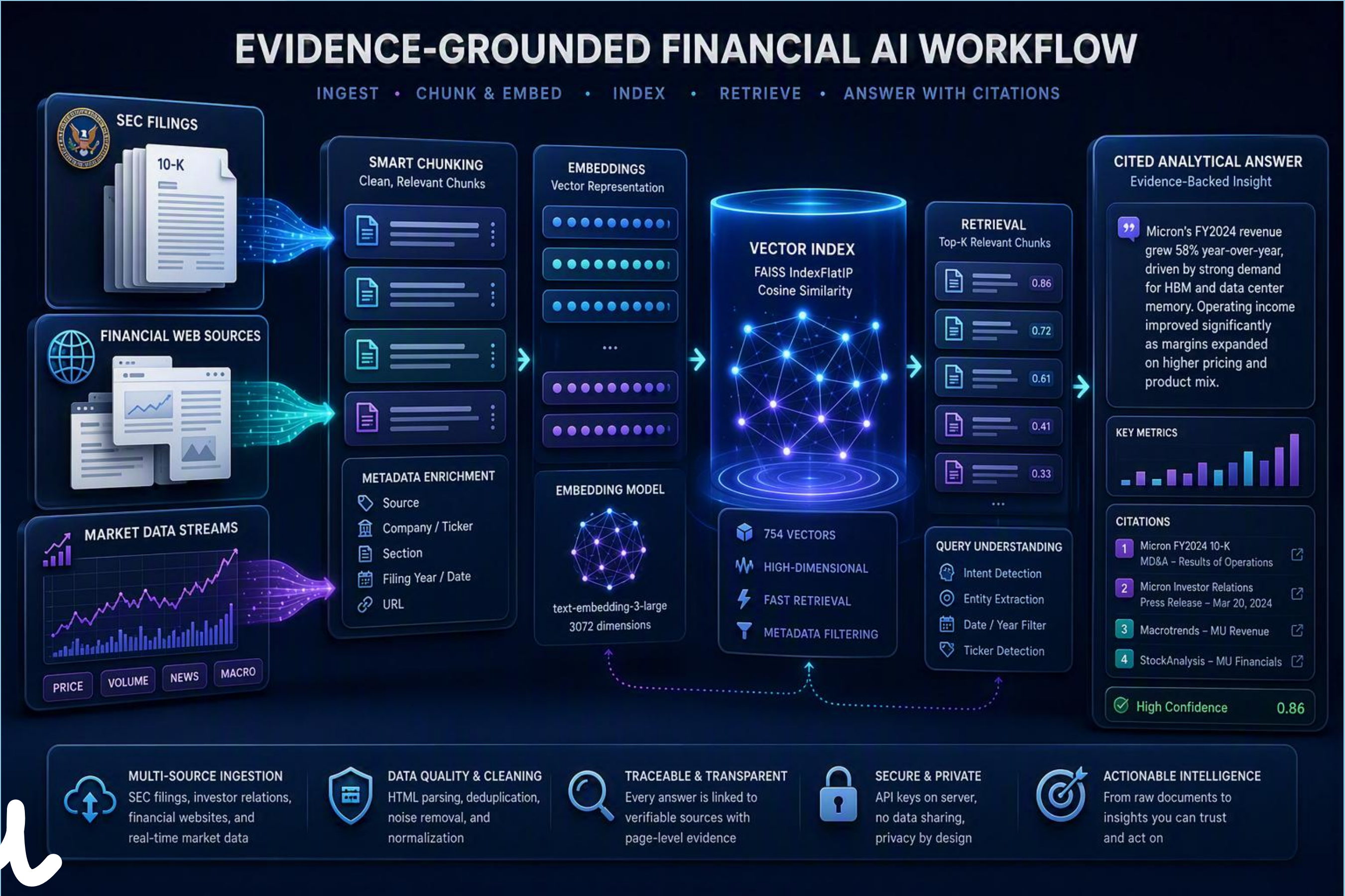
100-token overlap

Retrieval strategy

Top-8 chunks

Ticker, year and source filters

Answer = context + history + citations



Tool Use Registry: Core AI foundation

01	Working Prototype	DONE	Streamlit dashboard + Flask RAG + static HTML + one-click launch End-to-end flow from ingestion and indexing to cited answer and visualization.
02	RAG	DONE	FAISS IndexFlatIP + GPT-4o + text-embedding-3-large Top-8 cosine retrieval with ticker, source and fiscal-year filters plus citations.
03	Advanced Chunking	DONE	build_index.py tiktoken cl100k_base Paragraph-aware 400–800 token chunks, recursive sentence split and 100-token overlap.
04	Agent / LangGraph	DONE	rag_server.py orchestration + dashboard decision matrix Intent parsing, routing, state and quality × timing action generation.
05	Memory	DONE	Server sessions + localStorage + Chat History Up to 100 messages per session, 1-week TTL and conversational bypass of RAG.

VIVA message: the system is a complete application, not only an LLM prompt.

Tool Use Registry: Data, tools and sources

06	Tool Use	DONE	SEC EDGAR + Yahoo Finance + calculators + FAISS queries APIs, quantitative calculations and metadata-filtered vector search.
07	Web Search	DONE	Firecrawl + 21+ URLs + source-aware citations Six financial source types with URL, company, section and confidence metadata.
08	Local LLM	PARTIAL	Hybrid RAG path and GPT-4o general path Local Ollama integration remains planned for lower-cost general queries.
09	Data Pipeline	DONE	crawl → download → parse → clean → index → refresh Intermediate Markdown, HTML, CSV and JSON outputs support debugging and reruns.
10	Multiple Sources	DONE	SEC, Macrotrends, StockAnalysis, Yahoo, CMC, IR New sources can be appended to the SOURCES list with metadata fields.

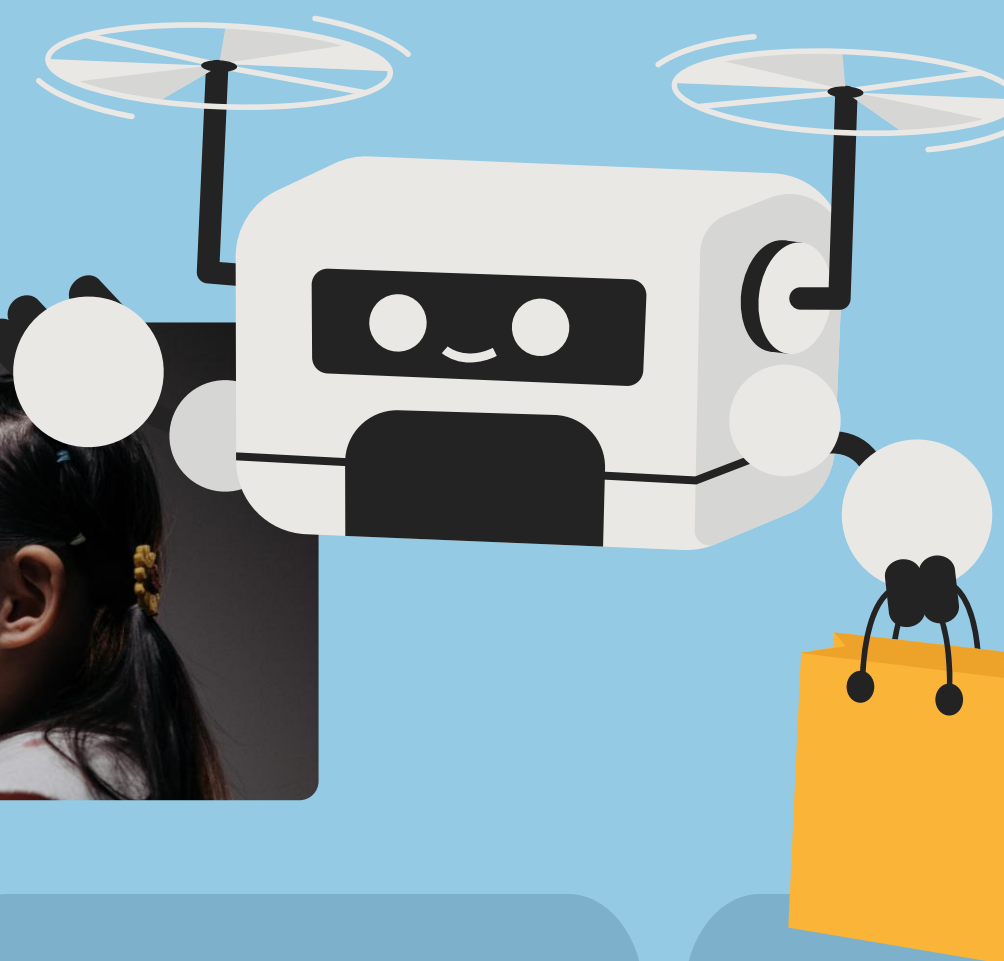
Key design choice: structured time-series stays in CSV; narrative evidence goes into FAISS.

Tool Use Registry: Evaluation, deployment and value

12	Evaluation	DONE	Confidence labels + failure handling + signal history	High ≥0.70, medium ≥0.40; retrieval failures are reported explicitly.
13	Deployment	DONE	Streamlit + Flask + static HTML + Windows launchers	Live dashboard, API-backed chat and browser-only report modes.
14	Bonus Tech	DONE	Plotly + ARIMA + 10,000-path Monte Carlo + quant signal	Interactive charts, forecasting ensemble and ten-factor timing engine.
15	Real Use Case	DONE	Intel and Micron semiconductor investment research	Actual filings and market history support comparison and decision analysis.
21	Financial web URLs			
6	Source types			
6	10-K filings			
754	FAISS vectors			
22k+	Stock CSV rows			

Honest assessment: 13 categories done, Local LLM partial, Fine-tuning intentionally not used.

Team Members



2028

Boon Ching

Core System

Working Prototype — End-to-end demo with Streamlit dashboard, Flask chatbot, static HTML.
RAG — Embedding + FAISS vector search + GPT-4o answers with citations.
Advanced Chunking — Smart document splitting with metadata.
Agent / LangGraph — Multi-step reasoning, decision matrix for investment actions.

Meridelle

Intelligence & Memory

Memory — Session memory, chat history, long-term FAISS persistence.
Tool Use — APIs (SEC EDGAR, Yahoo Finance), calculators, Monte Carlo, ARIMA.
Web Search — Firecrawl scraping 21+ financial sources with citations.
Local LLM (Hybrid) — Routing between RAG and GPT-4o fallback.

Jekan

Data & Sources

Data Pipeline — Ingestion, parsing, cleaning, ETL flow.
Multiple Data Sources — SEC EDGAR, Macrotrends, StockAnalysis, Yahoo Finance, CompaniesMarketCap, Investor Relations.
Fine-tuning / PEFT — Not used; relies on RAG + prompt engineering.
Evaluation — Retrieval confidence scoring, backtesting, failure handling.

Dorris

Deployment & Tech

Deployment — Streamlit, Flask API, static HTML, batch automation.
Bonus Tech — Plotly charts, ARIMA forecasting, Monte Carlo simulations, quant signals.
Bonus — Real Use Cases — Intel & Micron semiconductor analysis with real filings and market data.

Luo Ing

Environment & Libraries

Environment Variables — OPENAI_API_KEY, FIRECRAWL_API_KEY.
Libraries & Frameworks — OpenAI, FAISS, Tiktoken, Firecrawl, BeautifulSoup, yfinance, pandas, numpy, statsmodels, plotly, streamlit, flask, dotenv.



For Future Invest Today

makes semiconductor research
faster, traceable and more
decision-oriented.

Thank you

